Name: Score:

75 Multiple choice questions

Definition 1 of 75

B-picture

- Highest compression video frame
- "Pixel replication"
- "No reference to other pictures" / "Random access"
- Weighted Median with color images

Definition 2 of 75

Between frames uses motion compensation and requires a closed loop

- Intraframe coding
- Unsharp Masking Sequence
- MEQ VLC (Video Encoder Loop Sequence)
- Interframe coding

Definition 3 of 75

(1/MN) * sum([I(i j) - I_hat(i j)]^2)

- Average Optical Density (AOD) Formula
- Source Entropy Formula
- Mutual Information Formula
- Mean Square Error (MSE) Formula

Definition 4 of 75

Nearest Neighbor (Zero-order) Interpolation

- "Pixel replication"
- "backward prediction" / "future frame reference"
- "resolution reduction" / "format conversion"
- "unregistered planes" / "misregistration"

Definition 5 of 75

Pairs with index i

- Co-occurrence Matrix Row Mean (m_r)
- Co-occurrence Matrix Column Mean (m_c)
- Co-occurrence Matrix Total Sum (S)
- Co-occurrence Matrix Variance (var)

Definition 6 of 75

Must include scaling factor 1/N^2 to conserve energy

- "Continuous tone to 2 levels"
- "LSI" / "Invariant to space"
- Degradation Formula
- Inverse WHT Formula requirement

Definition 7 of 75

Artifact reduction or Compression

- Simple linear interpolation
- Traditional Huffman coding
- Basic thresholding algorithms
- Deep learning / CNN / GAN

Term 8 of 75

"Without considering spatial info" / "Solely by pixel intensity"

- Point Operations
- Geometrical Operations
- Global Operation
- Filters

Definition 9 of 75

(1 / (sqrt(2**pi)**sigma)) **e^(-(z-z_bar)^2 / (2**sigma^2))

- Source Entropy Formula
- Degradation Formula
- Coding Efficiency Formula
- Gaussian Noise PDF Formula

Definition 10 of 75

Median Cut Algorithm

- "Resolution reduction" / "Format conversion"
- "Pixel replication"
- "Color Palette Design"
- "Unregistered planes" / "Misregistration"

Definition 11 of 75

Wavelet Transform

- "spatial domain convolution" / "linear filtering"
- "frequency domain transformation" / "dc component removal"
- "single resolution analysis" / "fixed frequency band"
- "Multiresolution theory" / "Scaling and wavelet functions"

Definition 12 of 75

Must be non-zero along an intensity ramp (steady slope)

- Second-Order Derivative trigger
- Filter for Gaussian/White Noise
- Highest compression video frame
- First-Order Derivative trigger

Definition 13 of 75

Erosion

- "pixel replication"
- "unregistered planes" / "misregistration"
- "expands objects"
- "Shrinks objects"

Term 14 of 75

"Mathematical modeling directly from the image" / "Unknown point spread function"

- Blind Deconvolution
- Enhancement
- Notch Filter
- Halftoning

Definition 15 of 75

g(x y) = H[f(x y)] + n(x y) where noise is strictly additive

- Degradation Formula
- Source Entropy Formula
- Mean Square Error (MSE) Formula
- Coding Efficiency Formula

Definition 16 of 75

Gaussian Pyramid

- "Start processing at the lowest resolution"
- "start processing at the highest resolution"
- "combine all resolutions into one layer"
- "apply filters only to the edges"

Definition 17 of 75

Add four neighbors (up down left right) and subtract 4 times the center pixel

- Sobel Operator for horizontal edge
- Prewitt Operator for vertical edge
- Roberts Cross for diagonal edge
- Laplacian Calculation for center pixel

Definition 18 of 75

Target a fixed value of r

- "line tangent to the origin"
- "parabolic curve from the origin"
- "Circle centered on the origin"
- "radial direction from the origin"

Definition 19 of 75

Intra - Predictive - Bi-directional - DC

- I Prefer Small Cats (MPEG-2)
- I Prefer Big Dogs (MPEG-1)
- I Prefer Bright Lights (MPEG-4)
- I Prefer Fast Cars (MPEG-3)

Definition 20 of 75

Halftoning

- "full color to grayscale"
- "3d rendering to 2d images"
- "vector graphics to raster images"
- "Continuous tone to 2 levels"

Definition 21 of 75

Enhancement

- Focus on human preference or visual quality
- Calculate the total information entropy
- Eliminate noise or compression artifacts
- Measure the geometric distortion of pixels

Definition 22 of 75

Huffman (smallest) - Arithmetic (interval) - BTC (mean/var) - Vector Quantization (codebooks)

- **Happy Animals Bark Vigorously (Formats)**
- **Hungry Alligators Bite Viciously (Processes)**
- **Huge Animals Bite Vampires (Codecs)**
- **Hasty Ants Build Villages (Structures)**

Definition 23 of 75

Dirac Delta / Impulse Function

- **"periodic" / "repeating pattern"**
- **"LSI" / "Invariant to space"**
- **"non-linear" / "dependent on intensity"**
- **"gaussian" / "smooth curve"**

Definition 24 of 75

Dilation

- **"reduces dimensionality" / "loss of spatial info"**
- **"pixel replication"**
- **"shrinks objects"**
- **"Expands objects"**

Definition 25 of 75

Lossless Compression

- "lossy compression" / "high compression ratio"
- "Identical to original" / "No information lost"
- "significant data loss" / "approximation of original"
- "perceptual coding" / "human visual system model"

Definition 26 of 75

Harmonic Mean Filter (due to division by zero)

- Filter that enhances high-frequency details
- Filter that fails completely on Pepper noise
- Filter that works perfectly on Gaussian noise
- Filter that removes salt and Pepper noise

Term 27 of 75

"No reference to other pictures" / "Random access"

- B-picture
- Image Histogram
- I-pictures (Intra-coded)
- Block Truncation Coding (BTC)

Definition 28 of 75

Use the magnitude (absolute value) of the weights

- Arithmetic Mean with signed samples
- Geometric Mean with signed samples
- Weighted Median with signed samples
- Harmonic Mean with signed samples

Term 29 of 75

Focus on a priori knowledge or degradation models

- Quality
- Restoration
- Promotion
- Containment

Definition 30 of 75

Within the frame does not use motion compensation and uses an open switch

- Intraframe coding
- All or Nothing Rule (Graph Cuts)
- Interframe coding
- Linear Spatial Filters

Term 31 of 75

"Minimum mean square error" / "Uncorrelated noise"

- Wiener Filter
- Canny Edge Detector
- Wavelet Transform
- Notch Filter

Definition 32 of 75

Walsh-Hadamard Transform (WHT)

- "complex exponential basis" / "sine and cosine waves"
- "floating point coefficients" / "smooth waveforms"
- "gaussian kernel" / "continuous basis"
- "+1 and -1" / "Square waveforms"

Definition 33 of 75

Motion Estimation

- "spatial filtering" / "edge detection"
- "static image analysis" / "single frame processing"
- "color space conversion" / "pixel intensity mapping"
- "Rigid body motion" / "Differences between frames"

Definition 34 of 75

Discrete Hartley Transform (DHT)

- "Real-valued output" / "cas(x)"
- "pixel replication"
- "resolution reduction" / "format conversion"
- "color palette design"

Definition 35 of 75

Block Truncation Coding (BTC)

- "huffman coding" / "variable length codes"
- "run-length encoding" / "consecutive pixel repetition"
- "Sample mean and standard deviation"
- "wavelet transform" / "multiresolution analysis"

Definition 36 of 75

Pairs with index j

- Co-occurrence Matrix Standard Deviation (std)
- Co-occurrence Matrix Total Sum (S)
- Co-occurrence Matrix Row Mean (m_r)
- Co-occurrence Matrix Column Mean (m_c)

Definition 37 of 75

Harmonic Mean Filter

- "Bright white misfires" (Salt noise)
- "edge enhancement" (high-pass filtering)
- "dark black spots" (pepper noise)
- "gaussian blur effect" (low-pass filtering)

Definition 38 of 75

Morphological Filtering

- "statistical noise reduction" / "gaussian smoothing"
- "Union intersection complement" / "Geometrical structure"
- "pixel intensity scaling" / "color mapping"
- "fourier domain filtering" / "frequency response"

Definition 39 of 75

Observation - Experimentation - Mathematical Modeling

- MEQ VLC (Video Encoder Loop Sequence)
- OEM (Degradation Estimation Sequence)
- "Mathematical modeling directly from the image" / "Unknown point spread function"
- PSNR (Peak Signal-to-Noise Ratio) identification

Definition 40 of 75

H(S) = -sum(p_k * log_2(p_k))

- Structural Similarity Index (SSIM) Formula
- Peak Signal-to-Noise Ratio (PSNR) Formula
- Source Entropy Formula
- Mean Square Error (MSE) Formula

Definition 41 of 75

Motion Estimation - Quantization - Variable Length Coder

- Unsharp Masking Sequence
- PSNR (Peak Signal-to-Noise Ratio) identification
- OEM (Degradation Estimation Sequence)
- MEQ VLC (Video Encoder Loop Sequence)

Definition 42 of 75

Superpixels

- "uniform color regions"
- "geometric shapes in images"
- "Perceptually meaningful primitives" / "SLIC"
- "random pixel clusters"

Definition 43 of 75

Spatial-Spectral Transform

- "Registered planes"
- "unregistered planes" / "misregistration"
- "lsi" / "invariant to space"
- "color palette design"

Definition 44 of 75

Image Histogram

- "calculates the fourier transform" / "removes high frequencies"
- "applies a linear filter" / "changes color space"
- "preserves all spatial relationships" / "increases pixel count"
- "Reduces dimensionality" / "Loss of spatial info"

Definition 45 of 75

Discrete Cosine Transform (DCT)

- "wavelet decomposition" / "multiresolution analysis"
- "linear phase shift" / "complex coefficients"
- "odd symmetry" / "no dc component"
- "2N-point periodicity" / "Even symmetry"

Definition 46 of 75

Source (Background) - Sink (Foreground) - T-links (Connections)

- **SST (Graph Cuts)**
- **"No reference to other pictures" / "Random access"**
- **"Source and Sink terminal nodes"**
- **MEQ VLC (Video Encoder Loop Sequence)**

Definition 47 of 75

B-pictures (Bi-directional)

- **"Backward prediction" / "Future frame reference"**
- **"forward prediction" / "past frame reference"**
- **"static image analysis"**
- **"real-time rendering"**

Definition 48 of 75

Discrete Sine Transform (DST)

- **"No DC component" / "Odd symmetry"**
- **"Real-valued coefficients" / "Orthogonal basis"**
- **"Even Symmetry" / "DC component present"**
- **"Spatial Domain filtering" / "Convolution kernel"**

Definition 49 of 75

Look for 20 log10 and MSE in the denominator to the 1/2 power

- **PSNR (Peak Signal-to-Noise Ratio) identification**
- **SNR (Signal-to-Noise Ratio) identification**
- **RMSE (Root Mean Squared Error) identification**
- **MSE (Mean Squared Error) calculation**

Definition 50 of 75

Complex Spatial-Spectral Transform

- **"Unregistered planes" / "Misregistration"**
- **"lsi" / "invariant to space"**
- **"color palette design"**
- **"pixel replication"**

Definition 51 of 75

Graph Cuts

- **"Node connections and disconnections"**
- **"Input and Output data streams"**
- **"Source and Sink terminal nodes"**
- **"Graph vertices and edges"**

Definition 52 of 75

(1/MN) **sum(k** H_f(k))

- **Mean Square Error (MSE) Formula**
- **Source Entropy Formula**
- **Peak Signal-to-Noise Ratio (PSNR) Formula**
- **Average Optical Density (AOD) Formula**

Definition 53 of 75

Median Filter or Contraharmonic Mean

- **Filter for Color enhancement**
- **Filter for Salt and Pepper noise**
- **Filter for Gaussian blur**
- **Filter for Edge detection**

Term 54 of 75

"Periodic noise" / "Concentrated bursts of energy"

- **None Of The Mentioned**
- **Notch Filter**
- **Band Pass**
- **Unsharp Mask Filter**

Term 55 of 75

Filter for Periodic Noise

- None Of The Mentioned
- Notch Filter
- Band Pass Filter
- Unsharp Mask Filter

Definition 56 of 75

Image Restoration

- "A priori knowledge" / "Degradation model"
- "image segmentation" / "edge detection"
- "color space conversion" / "pixel manipulation"
- "feature extraction" / "object recognition"

Definition 57 of 75

Hartley - Cosine - Sine - Walsh-Hadamard - Wavelets

- Hats Can Shield Winds (Shadows)
- Horses Can Swim Well (Filters)
- Happy Cats Sing Whimsically (Patterns)
- Harry Can See Wavelets (Transforms)

Definition 58 of 75

Union equals entire image V and Intersection equals empty set

- All or Nothing Rule (Graph Cuts)
- MEQ VLC (Video Encoder Loop Sequence)
- Union equals intersection and Intersection equals union
- Intersection equals entire image V and Union equals empty set

Term 59 of 75

Filter for Gaussian/White Noise

- Pseudoinverse Filter
- Wiener Filter
- Notch Filter
- Harmonic Mean Filter

Term 60 of 75

"Nonblock codes" / "Interval of real numbers [0 to 1)"

- Huffman Coding
- Run-length Coding
- Arithmetic Coding
- Dictionary Coding

Definition 61 of 75

Target a fixed value of theta

- "line passing through the origin"
- "circle centered on the origin"
- "spiral outwards from the center"
- "Radial direction from the origin"

Definition 62 of 75

Sequential - Progressive - Complex - Hierarchical

- All or Nothing Rule (Graph Cuts)
- SPCH (JPEG Operational Modes)
- I Prefer Big Dogs (MPEG-1)
- MEQ VLC (Video Encoder Loop Sequence)

Definition 63 of 75

Encoded Bit Rate / Decoded Bit Rate

- Coding Efficiency Formula
- Source Entropy Formula
- Degradation Formula
- Gaussian Noise PDF Formula

Definition 64 of 75

Huffman Coding

- "spatial domain filtering" / "kernel convolution"
- "lossy compression technique" / "perceptual coding"
- "Smallest possible number of code symbols"
- "fixed-length code symbols" / "equal probability assumption"

Definition 65 of 75

Must be zero along an intensity ramp and only fires at the edges

- Filter for Gaussian/White Noise
- Highest compression video frame
- First-Order Derivative trigger
- Second-Order Derivative trigger

Definition 66 of 75

Box - Gaussian - Laplacian

- "Registered Planes"
- "No Reference To Other Pictures" / "Random Access"
- Non-linear Filters
- Linear Spatial Filters

Definition 67 of 75

Video Transcoding

- "frame rate increase"
- "Resolution reduction" / "Format conversion"
- "quality enhancement" / "color adjustment"
- "compression without conversion"

Definition 68 of 75

Ideal Low-Pass Filter

- "removing dc component" / "odd symmetry"
- "Forcing frequency response to zero"
- "amplifying high-frequency noise" / "sharpening edges"
- "preserving all frequency components" / "identity transformation"

Definition 69 of 75

Upsampling / Interpolation

- "rigid body motion" / "differences between frames"
- "no dc component" / "odd symmetry"
- "Insert zeros between samples"
- "no reference to other pictures" / "random access"

Definition 70 of 75

Blur then Subtract to get mask then Add

- Filter For Gaussian/White Noise
- Unsharp Masking Sequence
- MEQ VLC (Video Encoder Loop Sequence)
- Linear Spatial Filters

Definition 71 of 75

Vector weighted median filter

- Weighted Median with color images
- Harmonic Mean with color images
- Geometric Mean with color images
- Arithmetic Mean with color images

Definition 72 of 75

Opening is Erosion then Dilation

- CDE (Morphological Mastery)
- Filter for Gaussian/White Noise
- "Expands objects"
- OED (Morphological Mastery)

Definition 73 of 75

Closing is Dilation then Erosion

- "Expands objects"
- "No DC component" / "Odd symmetry"
- OED (Morphological Mastery)
- CDE (Morphological Mastery)

Definition 74 of 75

Median - Erosion - Dilation

- Linear Spatial Filters
- Non-linear Filters
- Filter For Salt And Pepper Noise
- Filter For Gaussian/White Noise

Term 75 of 75

"Salt-and-Pepper noise" / "Film-grain noise" / "Preserve edges"

- Notch Filter
- Median Filter (Nonlinear)
- Block Truncation Coding (BTC)
- Ideal Low-Pass Filter